

Robotics Club of BRAC University

A story of curiosity, courage, and the machines that moved us forward.

ROBU: The Heartbeat of Robotics Dream

By **Ayesha Sneha**



ROBU community at Prototype Summer 2026 · Photo by ROBU

From that vision, a legacy began.

Imagine a place filled with wires, wheels, and exciting ideas, with a soft haze of soldering smoke in the air and everything drawing the mind toward invention. Every BRAC University student would recognize this place as the Robotics Club of BRAC University, where imagination becomes reality.

Let's look at ROBU's journey. Every semester, the club recruits members through a competitive process. Students take part in one-to-one interviews with senior members, learning more about the club and how it works. After selection, they attend a Basics of Robotics workshop. The club believes every student should have the opportunity to experience engineering firsthand. Young minds build their first circuits and write their first lines of robotic code. In those moments, they begin to hold their future dreams.

Every great story needs an even greater guide. Dr. Md. Khalilur Rahman began working at BRAC University in 2009. He earned a Ph.D. in Natural Language Processing from Kyushu Institute of Technology in Japan. His interests extended beyond books, with expertise in artificial intelligence and natural language processing. With one goal in mind, he explored robotics, embedded systems, and space technology, giving students access to real-world platforms. The BRAC University Robotics Lab was founded as a place where concepts could be tested, refined, and transformed.

In 2011, ROBU made headlines with ChondroBot, Bangladesh's first lunar excavation robot, competing in NASA's prestigious Lunabotics Mining Competition. A year later, ChondroBot-2 returned to NASA's competition. It won Asia's Best, secured 12th place internationally, and received the Joe Kosmo Award of Excellence.

Then came Mongol-Tori, ROBU's Mars rover. First presented at the University Rover Challenge (URC) in 2018, it ranked 13th among 95 teams worldwide. The team kept pushing forward: Mongol-Tori placed 4th in URC 2021, became Asia's second-best in 2023, and reached 8th globally in 2025. It also won the AUST Rover Challenge, excelled at the Digital World ICT Expo, and earned recognition from the Space Innovation Summit and the National ICT Awards.

In 2017, BRACU Onnesha, Bangladesh's first nanosatellite, was launched as part of a global project with Kyushu Institute of Technology. On June 3, 2017, SpaceX carried it to the International Space Station aboard the CRS-11 mission. It proudly carried Bangladesh's flag.

The momentum continued. In 2018, ROBU introduced BRACU Duburi, the country's first autonomous underwater vehicle (AUV), which competed at the International RoboSub Competition. By 2023 and 2024, innovations such as Ognibir, a firefighting rescue robot, and Nabik, designed for flood relief, showed how robotics could serve humanity in times of crisis.

And the story is still unfolding. In February 2025, ROBU launched "Traction অভ্যুদয়," a national robotics contest presented by the OPPO Reno13 Series. The event brought young innovators from across Bangladesh together to think beyond their limits.

From NASA to Mars, from satellites to submarines, ROBU is more than a university club. It is a testament to what young Bangladeshis can build when courage is met with support. ROBU is where curiosity leads to creation, and where the impossible becomes possible.

Codex event pulled off successfully

By Humayra Saniat Lithe



Photo posted by Shahriyar Al Mustakim Mitul

Three hours of hands-on exploration turned ideas into working prototypes—and brought innovation, collaboration, and learning together.

Codex: BuildCave successfully brought together innovation, collaboration, and hands-on learning in an engaging three-hour interactive experience, powered by **Microsoft Build** and **Qwen**.

The session began with an immersive Microsoft Build hands-on lab, where participants explored Microsoft's development ecosystem through guided practical exercises. This was followed by an exciting agentic AI-driven development session where participants experienced the power of AI-assisted development and transformed ideas into working prototypes using the provided API credits.

Throughout the event, participants actively engaged in building, experimenting, and collaborating, creating an energetic learning environment filled with curiosity and creativity. The enthusiasm and dedication shown by everyone made the session both productive and inspiring.

The event concluded on a high note as participants showcased their builds, with the top projects earning **\$100 Codex API Credits** in recognition of their creativity and technical execution.

A huge thank you to everyone who joined us and contributed to making **Codex: BuildCave** a successful and memorable experience. We also extend our sincere gratitude to **Microsoft Build** and **Qwen** for powering the event and supporting hands-on AI learning.

Here's to building more, learning more, and innovating together.

Broadband Expo held successfully by ROBU

By Nabil Wasit



Photo by ROBU

More than a showcase, the expo reflected an innovation ecosystem in motion—and the student teams bringing it to life.

At the Broadband Expo hosted by the **ISP Association of Bangladesh**, a quiet but compelling story of innovation unfolded over three days—one that spoke not only of machines and prototypes, but of persistence, vision, and a growing national confidence in homegrown technology. The **Robotics Club of BRAC University** successfully showcased its talents and outstanding projects to the audience.

The initiative to participate in the expo was undertaken by ROBU, whose members curated and presented a cross-section of one of the country's longest-running student-driven robotics ecosystems. What they brought to the exhibition floor was not just a display, but a narrative built over 15 years. They have been constantly surpassing their limits to generate innovative ideas and exceptional performance.

A major feature of the event was the presence of **Zahir Uddin Swapon**, Honorable Minister of Information and Broadcasting, alongside **Rehan Asif Asad**, Advisor to the Prime Minister. Their visit added both recognition and encouragement to what is, at its core, a student-led journey. It underscored a growing alignment between policy, education, and innovation. The honorable minister highly praised and appreciated the projects undertaken and made by passionate young students of ROBU.

At the heart of the showcase was BRAC University's robotics ecosystem, an effort that has evolved steadily through years of experimentation, mentorship, and relentless effort. What began as a modest initiative has matured into a platform that not only builds machines but also nurtures talent and ideas. Here, ROBU proved that it can accomplish the best with its remarkable talents and hard work.

Autonomous and semi-autonomous systems such as **BRACU Duburi**, **BRACU Mongol-Tori**, and **BRACU Ognibir** drew significant attention, alongside service-oriented restaurant robots that demonstrated practical, everyday applications of robotics. Each system reflected a different approach to solving real-world challenges, from exploration to automation. They all

serve an important purpose in the technological aspects of Bangladesh's IT sectors.

For many visitors, the experience was unexpectedly revealing. A recurring sentiment echoed across conversations: surprise. There remains a widespread perception that cutting-edge robotics belongs elsewhere. The exhibition challenged that assumption. Here was clear evidence that advanced research and development is taking place within Bangladesh, driven by students who are not only learning but actively contributing to technological progress.

By the third day, the exhibition settled into a more conversational rhythm. Without the formality of high-profile visits, deeper discussions began to take shape. Industry representatives, academics, and visitors engaged closely with the teams, leading to multiple expressions of interest in collaboration. These interactions pointed toward a promising bridge between academic innovation and industry application.

What emerged over the three days was more than a showcase; it was a reflection of an ecosystem in motion. Behind each project stands a network of dedicated individuals who have invested years of effort, often working beyond formal hours, to bring ideas into reality. The impact of this work is already visible, with startups emerging from the ecosystem and its influence continuing to expand. ROBU ensured its work was highly evaluated and checked by the best critics.

Yet perhaps the most important takeaway lies in visibility. For all its progress, much of this work remains underrepresented in public discourse. Platforms like the Broadband Expo offer a crucial opportunity to tell these stories, make visible what is being built, and inspire a wider audience. This event helped indirectly advertise the work and created a medium to interact face-to-face with policymakers, elected officials, and the general public.

There is a growing sense that such initiatives should not be occasional, but regular. Showcasing local innovation, fostering dialogue, and connecting ecosystems with industry are not just beneficial; they are necessary. If Bangladesh is to fully realize its technological potential, it must continue to create and support more spaces where ideas can be shared openly and achievements can be seen and understood.

In this sense, the ROBU effort went beyond simply attending an expo. It served as a reminder that the nation's innovation story is already being written and should be shared more frequently and more loudly.

COMPETITION FEATURE · GLOBAL ACHIEVEMENT



Source: TBS Graduates

Duburi secures 8th position globally in ROBOSUB

The **Robotics Club of BRAC University** congratulates **BRACU Duburi** for securing 8th position globally and 2nd in Asia at ROBOSUB 2025. The team also represented Bangladesh with strong results in the website and introductory video assessments.

The achievement continues a journey of homegrown underwater robotics and inspires young innovators. Dubotech, a startup founded by team alumni, carries that work into industry.

Innovation that keeps moving forward

DuboMini, the Duburi team's latest creation, is an assistant UAV to the enhanced **Duburi 4.2**. It took the stage with its inter-bot signal transmission, demonstrating coordinated communication in the underwater arena. With remarkable efficiency, the achievement felt like capturing the moment through a fish-eye lens.

We once again congratulate the entire DUBURI team for overcoming every obstacle and leading the way for Bangladesh on the global stage. It is an absolute honour to recognize their work, effort, and outstanding performance in representing our country, our university, and all of us.

BRACU Ognibir wins Mindsparks Project Showcasing

By Mohammed Tashfiqul Islam



Photo by ROBU

A first-of-its-kind firefighting robot brings autonomy, rescue, and civil safety together.

Heartiest congratulations to **BRACU Ognibir** for winning the AUST Mindsparks Project Showcasing with the country's first firefighting robot: ROBU's first semi-autonomous firefighting vehicle. This latest project by BRACU and ROBU was undertaken as an initial effort to integrate autonomy, firefighting, and rescue robotics—and to demonstrate a new approach to civil rescue work using robots.

The semi-autonomous bot was designed to work as an industrial-grade machine and civil rescue robot in fire emergencies. The Ognibir team's expertise, effort, and hard work brought its unique concepts to life and secured an impressive victory against strong competitors.

ROBU will continue striving to create extraordinary work with professionalism and skill. Once again, congratulations and applause to the BRACU Ognibir team for this meritorious triumph.

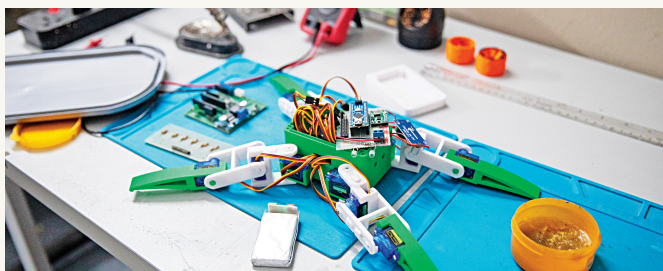
BANGLADESH ROBOTICS ECOSYSTEM

Building what comes next

Three homegrown teams are bringing robotics into classrooms, workplaces, and real-world operations.

Writeups by **Zasheea Nazah Rameesa**

BOT Engineers



Md Ali Razin with colleagues at BOT Engineers · The Daily Star

BOT Engineers is a robotics ecosystem built around three pillars: BOT Academy, BOT Café, and BOT Solution, making robotics more accessible through community, education, and innovation.

BOT Café, Bangladesh's first robotics co-working space, provides workstations, soldering tools, and maker equipment for robotics enthusiasts and engineers. Through BOT Academy, students aged 8–16 gain hands-on robotics experience through engaging, real-world learning in collaboration with schools.

BOT Solution designs and develops custom robots for businesses, events, and personal projects, including drones, battle robots, automation, and self-driving systems. With years of practical experience and participation in international robotics competitions, BOT Engineers brings strong technical expertise and rigorous testing to every project.

The team also contributes to the next generation of roboticists through mentorship, including guiding Mission Matrix, whose team went on to win eight gold medals across competitions in Malaysia and Indonesia.

Dubotech



Dubotech Digital Ltd wins National SME Entrepreneur Award 2025

Dubotech is an innovative robotics and autonomous systems company connecting vehicles and sensors to support security, inspection, and logistics across sea, air, and land.

Its platforms are purpose-built for operations above, across, and below the surface, with a shared-autonomy architecture that connects them through FORTIS. The company's six core systems include VTOL Drone, ROV, UGV, UUV, USV, and Fixed Assets, each designed for a specific operational mission.

From rapid disaster-response deliveries and underwater inspection to threat detection, autonomous patrol, and persistent surveillance, these systems address complex real-world challenges. Dubotech has also gained recognition through initiatives such as Mechnoverse – Beyond Gears and Gravity, Startup Connect 2025, and Shark Tank Bangladesh.

Through its journey from Duburi to Dubotech, the company continues to demonstrate how homegrown robotics can redefine possibilities across land, sea, and air.

Roboway Technologies



Pixi Bot · Roboway Technologies

Roboway Technologies is a robotics and AI firm developing intelligent systems across humanoid robotics, autonomous vehicles, drones, and IoT. The company combines robotics, artificial intelligence, and embedded systems to create practical solutions for real-world challenges.

Its innovative projects include PIXI Humanoid, TANVIN MINI 1.0 VTOL Surveillance Drone, DETECT X, AQUA BOT Autonomous Underwater Vehicle, and i-Care. These projects demonstrate applications ranging from intelligent human-machine interaction and aerial surveillance to underwater exploration and AI-powered healthcare.

Through advanced technologies such as computer vision, ROS2, autonomous control, and AI-based detection, Roboway develops systems designed for real-world deployment. With a strong focus on research and innovation, the company aims to push the boundaries of human-machine collaboration.